

Project Atomic-Link

Bloc 3: Real-Time Data Pipelines

Context: AuroraTech
Hardware Supply Chain



Financial Data



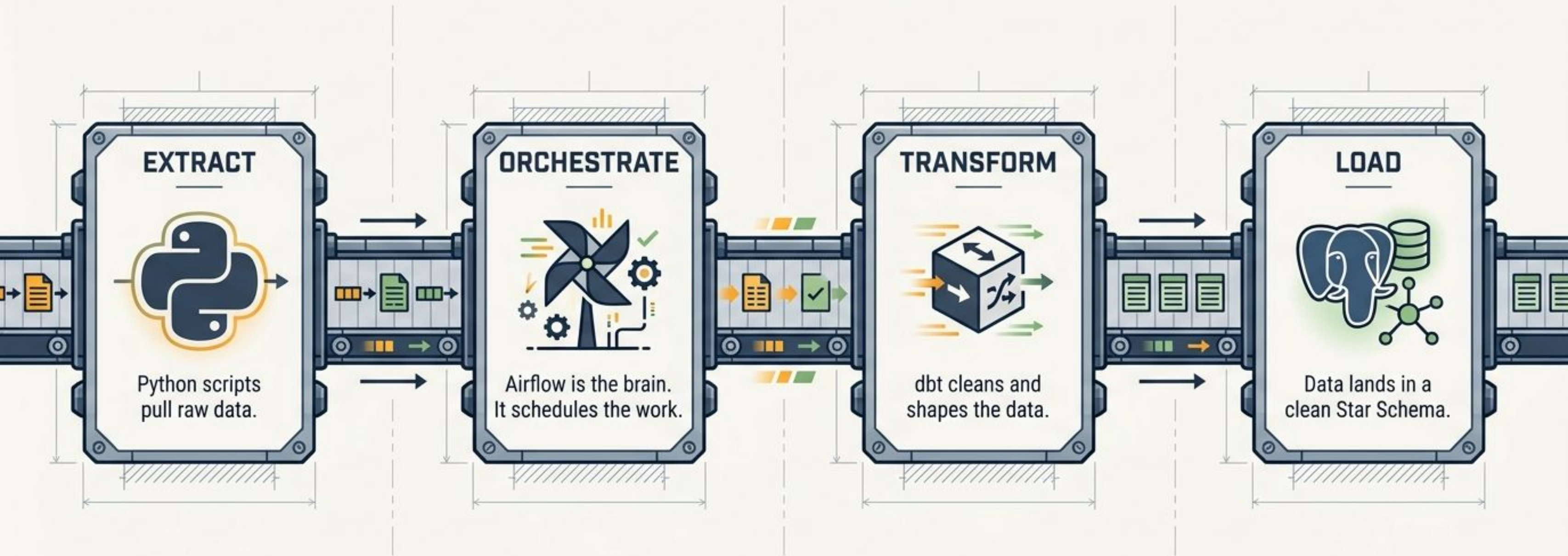
Logistics Data



Margin Protection

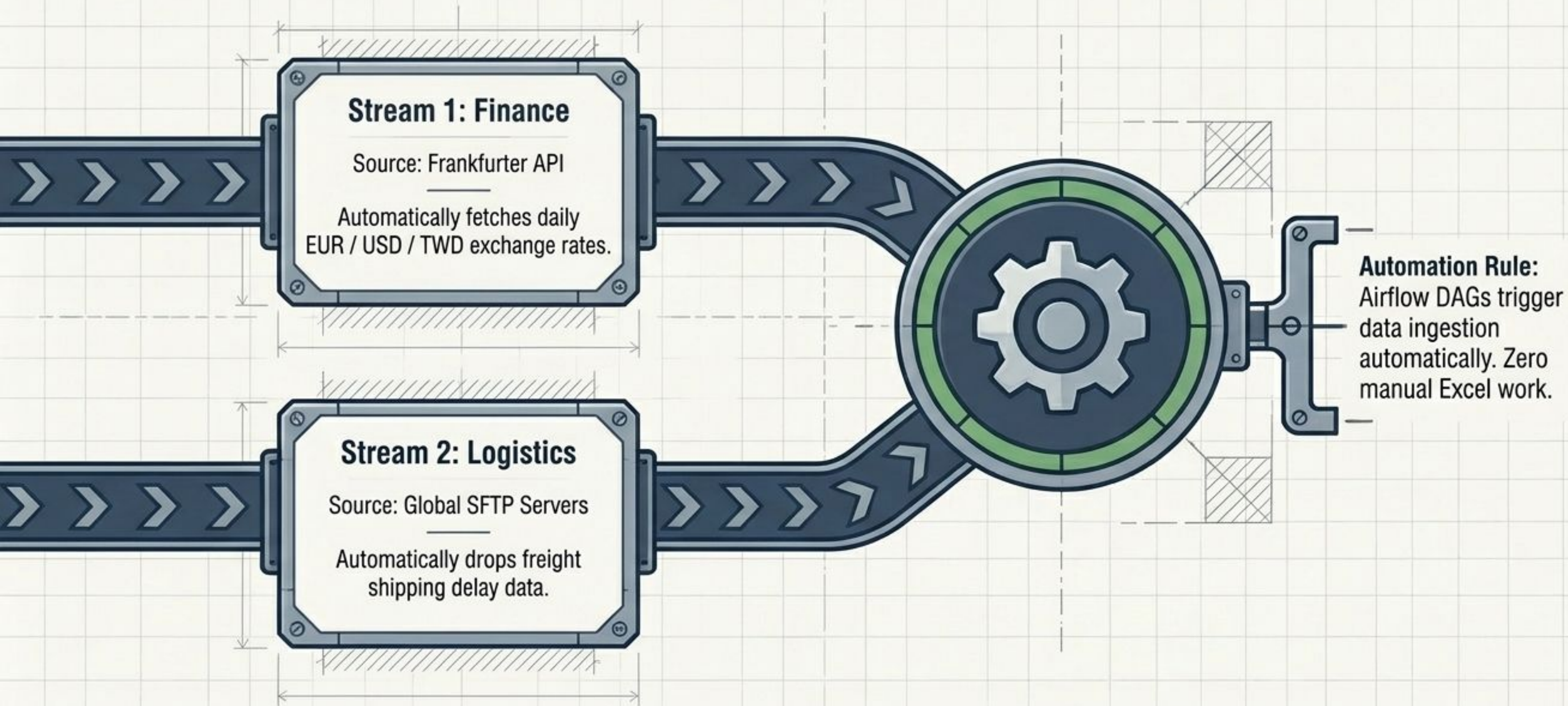
Business Goal: Unify real-time currency rates and shipping data into a Single Source of Truth (SSOT). We predict and protect company profit.

The Factory Floor: Pipeline Architecture

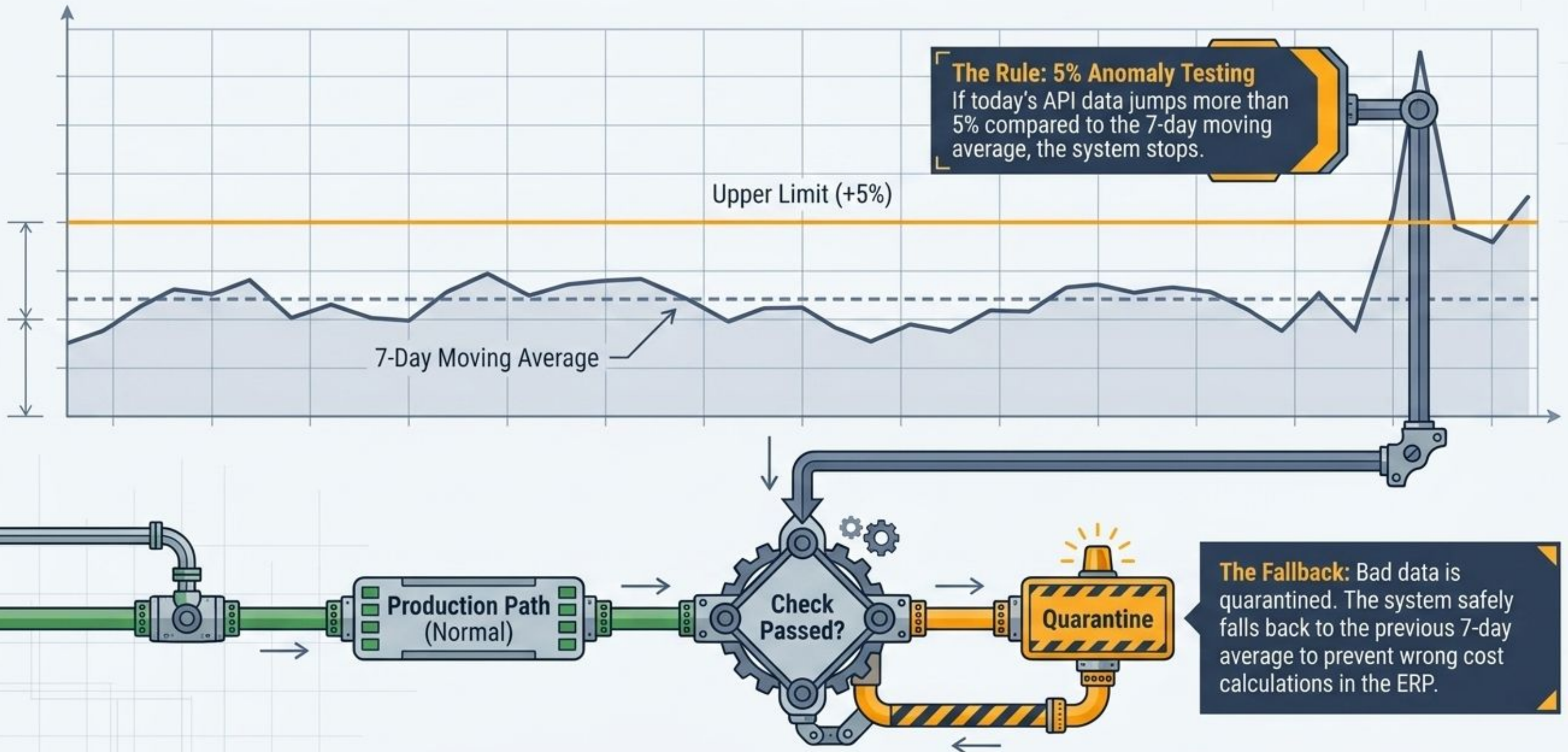


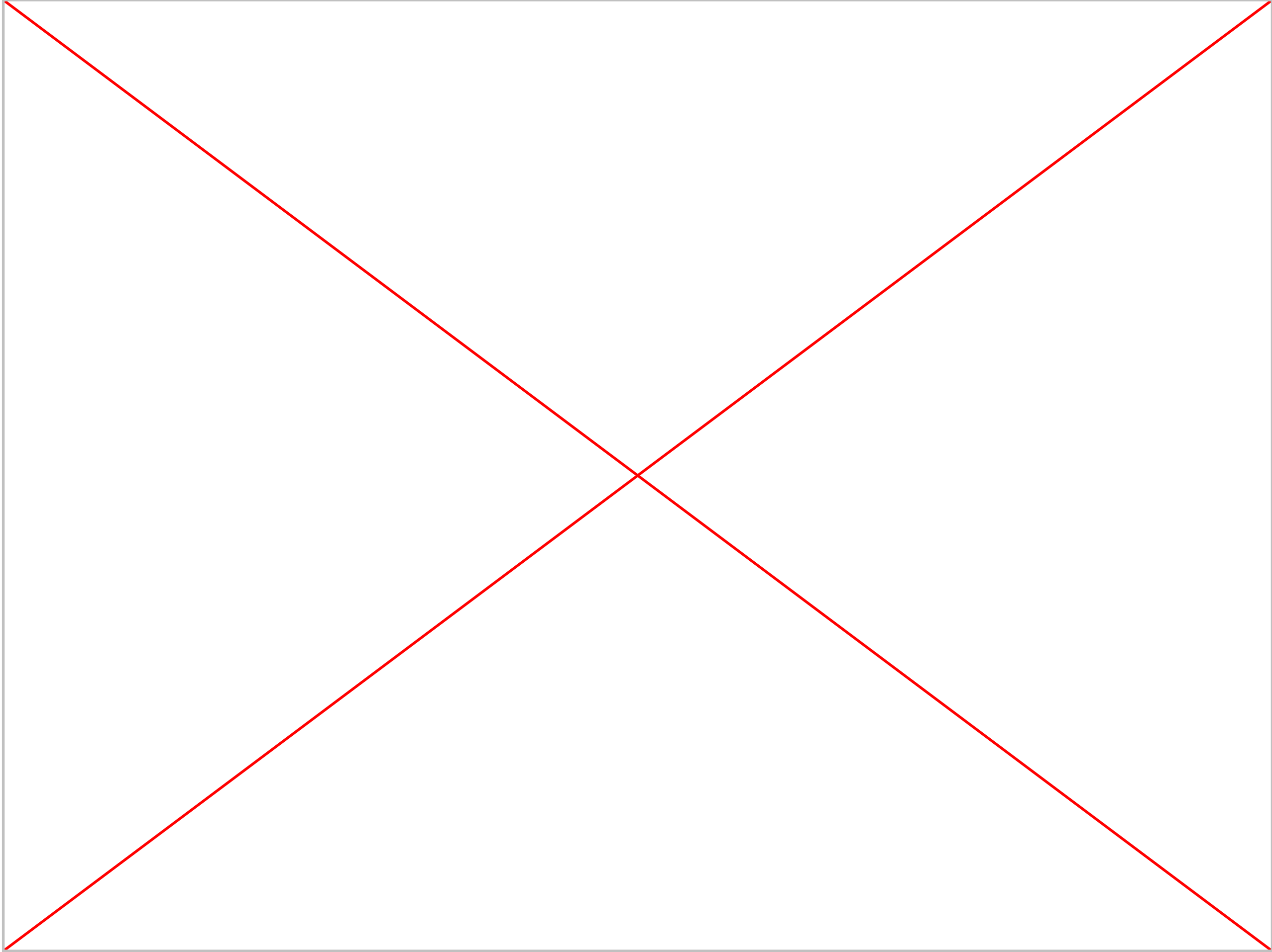
Airflow schedules the work. Python gets the data. dbt cleans and structures it.

The Conveyor Belt: Automated Ingestion



Quality Assurance: Error Handling







Simulate: TW Earthquake



DAG
Map

Runner
Metrics

Reset



Completed



Extract FX & Transit

✓ External Signals Ready



Predict Delays

⚠ Disruption Detected



Action Engine

✓ Mitigation Strategy Applied

LIVE DAG LOGS



UTC System Time

[INFO] Starting DAG execution
using scenario: earthquake

[OK] API: Fetched FX rates
(EUR/USD = 1.085)

[WARN] ALERT: Magnitude 6.5
Earthquake detected in Taiwan
region.

[ACTION] Hsinchu Science Park
facilities halted for safety
checks. 7-day delay.

[ACTION] Re-balancing inventory
from US reserves.

[OK] Data Warehouse fact tables
updated.

[SUCCESS] DAG Complete.



Dag
auroratech_realtime_pipeline

🏠
Home

🔗
Dags

🗄️
Assets

🌐
Browse

⚙️
Admin

📖
Docs

🕒
02:00

👤
User

🔍

⚙️

📄

📄

00:08:33

00:04:16

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▶️

▶️

extract_currency_rates

extract_shipping_logistics

transform_and_load

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🔗 auroratech_realtime_pipeline ☐

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Schedule

Latest Run

Next Run

Max Active Runs

Owner

Tags

📅 1 day, 0:00:00

2026-06-14 13:18:43 ✅

2026-06-15 13:17:58

0 of 16

data_engineering

🏷️ auroratech

Latest Dag Version

v1

Overview

Runs

Tasks

Calendar

Backfills

Audit Log

Code

Details

+ Filter

Dag Run ID	Run After	State	Run Type	Triggering User	Start Date	Duration
> manual__2026-06-14T11:18:43.232937+00:...	2026-06-14 13:18:43	✅ Success	▶ Manual	admin	2026-06-14 13:18:45	00:0
manual__2026-06-14T11:18:26.623136+00:...	2026-06-14 13:18:26	✅ Success	▶ Manual	admin	2026-06-14 13:18:28	00:0
scheduled__2026-06-14T11:17:58.699234+00:...	2026-06-14 13:17:58	✅ Success	🕒 Scheduled		2026-06-14 13:18:15	00:0

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The Control Room: Production Monitoring



**Result: Zero silent failures.
We fix issues before business
reports are affected.**